VTPEH 6270 – Check Point 06: Statistical Analyses

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Contents

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1 Introduction

#Link to GitHub: https://github.com/lydiamichael03/datanalysisR.git

1.1 Research Question

Does gonorrhea incidence rate differ significantly between males and females across age groups in the United States (2000–2022)?

1.2 Scientific Plausibility and Context

Gonorrhea (*Neisseria gonorrhoeae*) is one of the most commonly reported sexually transmitted infections (STIs) in the United States, with over 700,000 new cases estimated annually [@cdc2023]. The epidemiology of gonorrhea is not uniform across the population: age and sex are consistently identified as strong determinants of infection risk.

Adolescents and young adults (ages 15–24) bear a disproportionate burden of gonorrhea infection due to a combination of behavioral, biological, and social factors [@leichliter2007]. Biologically, the cervical ectopy present in adolescent females increases susceptibility to infection [@haggerty2010]. Behaviorally, younger individuals are more likely to have multiple partners and less likely to use barrier contraception consistently. However, rates among males have risen in recent decades, partly driven by increases in gonorrhea among men who have sex with men (MSM) [@cdc2023].

Understanding sex-specific patterns across age groups is critical for targeted public health interventions, including screening guidelines, partner notification programs, and treatment resource allocation [@leichliter2007].

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2 Material & Methods

2.1 Data

The dataset used in this analysis was obtained from the CDC's *National Notifiable Diseases Surveillance System* (NNDSS) and summarized via the CDC STI Surveillance reports (available at https://www.cdc.gov/std/statistics). Data were collected at 3-year intervals from 2000 to 2022 (years: 2000, 2003, 2006, 2009, 2012, 2015, 2018, 2020, 2022) and stratified by **sex** (Male, Female) and **age group** (15–19, 20–24, 25–29, 30–34, 35–44, 45–54, 55–64, 65+).

The dataset (`gonorrhea_clean.csv`) contains 144 observations and 7 variables:

Variable	Type	Description
`year`	Integer	Year of observation
`sex`	Character	Sex (Male / Female)
`age_group`	Character	Age group (8 categories)
`cases`	Integer	Number of reported gonorrhea cases
`rate`	Numeric	Rate per 100,000 population
`population`	Integer	Estimated population size
`rate_log`	Numeric	$\log_{10}$-transformed rate (pre-computed)

Data cleaning and transformation: The rate variable was $\log_{10}$-transformed (already included as `rate_log`) to address right skew in the distribution of rates, which is typical for count-based epidemiological data. The age group variable was converted to an ordered factor to preserve biological ordering. No missing values were present.

```r
library(tidyverse)
library(ggplot2)
library(car)
library(emmeans)
```

```r
# Load data
gonorrhea <- read.csv("gonorrhea_clean.csv", stringsAsFactors = FALSE)

# Factor ordering
gonorrhea$age_group <- factor(gonorrhea$age_group,
  levels = c("15-19","20-24","25-29","30-34","35-44","45-54","55-64","65+"),
  ordered = TRUE
)
gonorrhea$sex <- factor(gonorrhea$sex)
gonorrhea$year <- factor(gonorrhea$year)

# Quick summary
str(gonorrhea)
```

```
## 'data.frame':	144 obs. of  7 variables:
##  $ year      : Factor w/ 9 levels "2000","2003",..: 1 1 1 1 1 1 1 1 1 1 ...
##  $ sex       : Factor w/ 2 levels "Female","Male": 2 2 2 2 2 2 2 2 1 1 ...
##  $ age_group : Ord.factor w/ 8 levels "15-19"<"20-24"<..: 1 2 3 4 5 6 7 8 1 2 ...
##  $ cases     : int  61542 89201 64832 41021 34112 8823 2011 891 105321 91032 ...
##  $ rate      : num  313.2 448.7 320.1 192.4 87.6 ...
##  $ population: int  19648000 19882000 20257000 21320000 38920000 32188000 22596000 28792000 19762000
##  $ rate_log  : num  2.5 2.65 2.51 2.28 1.94 ...
```

```r
summary(gonorrhea[, c("sex","age_group","rate","rate_log")])
```

```
##      sex       age_group       rate           rate_log
##  Female:72   15-19   :18   Min.   :  1.00   Min.   :0.0004341
##  Male  :72   20-24   :18   1st Qu.: 15.43   1st Qu.:1.1882378
##              25-29   :18   Median :130.65   Median :2.1160364
##              30-34   :18   Mean   :183.98   Mean   :1.8147701
##              35-44   :18   3rd Qu.:331.35   3rd Qu.:2.5201569
##              45-54   :18   Max.   :585.90   Max.   :2.7678242
##              (Other):36
```

2.2 Statistical Analyses

To examine whether gonorrhea incidence rates differ by sex and age group, a **two-way ANOVA** was conducted using $\log_{10}$-transformed rate as the outcome variable, with `sex` and `age_group` as fixed factors, including their interaction term.

2.2.1 Assumption Checks

Prior to model fitting, the following assumptions were verified:

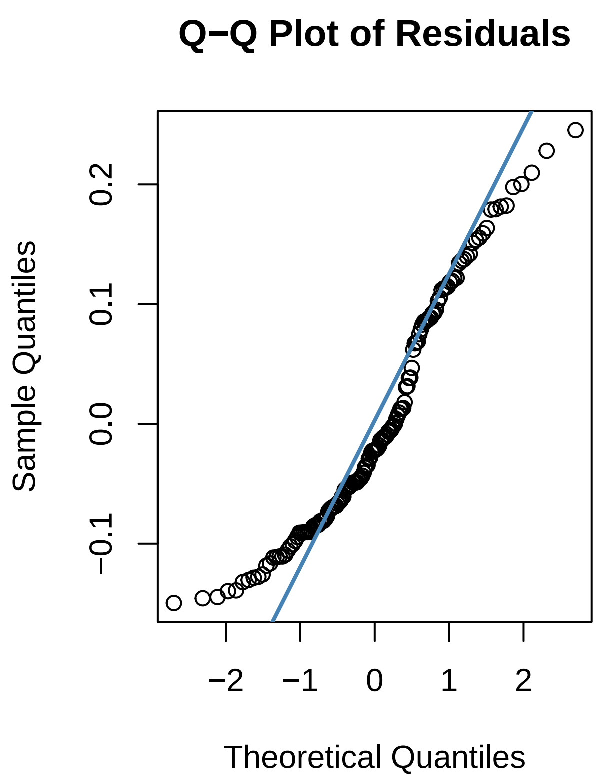
1. **Normality of residuals** – assessed via Shapiro-Wilk test and Q-Q plot.
2. **Homogeneity of variance** – assessed via Levene's test.
3. **Independence** – observations are from different sex × age group × year combinations; independence is assumed by study design.

```r
# Fit the full two-way ANOVA model
model <- aov(rate_log ~ sex * age_group, data = gonorrhea)

# --- Normality of residuals ---
shapiro.test(residuals(model))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(model)
## W = 0.94046, p-value = 8.553e-06
```

```r
# Q-Q plot of residuals
par(mfrow = c(1,2))
qqnorm(residuals(model), main = "Q-Q Plot of Residuals")
qqline(residuals(model), col = "steelblue", lwd = 2)

# --- Homogeneity of variance ---
leveneTest(rate_log ~ sex * age_group, data = gonorrhea)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value Pr(>F)
## group  15  1.2334 0.2555
##       128
```



2.2.2 Two-Way ANOVA

```r
summary(model)
```

```
##                 Df Sum Sq Mean Sq F value Pr(>F)    
## sex              1   1.32   1.321  124.12 <2e-16 ***
## age_group        7  90.56  12.937 1215.96 <2e-16 ***
## sex:age_group    7   1.69   0.241   22.69 <2e-16 ***
## Residuals      128   1.36   0.011                   
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

2.2.3 Post-Hoc Analyses

Given a significant interaction or main effect, pairwise comparisons between age groups within each sex were performed using the `emmeans` package with Tukey correction for multiple comparisons.

```r
# Estimated marginal means for sex × age_group interaction
emm <- emmeans(model, ~ sex * age_group)
pairs(emm, by = "sex", adjust = "tukey")
```

```
## sex = Female:
##  contrast          estimate     SE  df t.ratio p.value
##  (15-19) - (20-24)  -0.0531 0.0486 128  -1.092  0.9574
##  (15-19) - (25-29)   0.1452 0.0486 128   2.987  0.0647
##  (15-19) - (30-34)   0.4197 0.0486 128   8.632 <0.0001
##  (15-19) - (35-44)   0.8498 0.0486 128  17.477 <0.0001
##  (15-19) - (45-54)   1.3503 0.0486 128  27.771 <0.0001
##  (15-19) - (55-64)   1.8606 0.0486 128  38.264 <0.0001
##  (15-19) - (65+)     2.5140 0.0486 128  51.703 <0.0001
##  (20-24) - (25-29)   0.1983 0.0486 128   4.079  0.0020
##  (20-24) - (30-34)   0.4729 0.0486 128   9.725 <0.0001
##  (20-24) - (35-44)   0.9029 0.0486 128  18.569 <0.0001
##  (20-24) - (45-54)   1.4034 0.0486 128  28.863 <0.0001
##  (20-24) - (55-64)   1.9137 0.0486 128  39.357 <0.0001
##  (20-24) - (65+)     2.5671 0.0486 128  52.796 <0.0001
##  (25-29) - (30-34)   0.2745 0.0486 128   5.646 <0.0001
##  (25-29) - (35-44)   0.7046 0.0486 128  14.490 <0.0001
##  (25-29) - (45-54)   1.2051 0.0486 128  24.784 <0.0001
##  (25-29) - (55-64)   1.7154 0.0486 128  35.278 <0.0001
##  (25-29) - (65+)     2.3688 0.0486 128  48.717 <0.0001
##  (30-34) - (35-44)   0.4300 0.0486 128   8.844 <0.0001
##  (30-34) - (45-54)   0.9306 0.0486 128  19.138 <0.0001
##  (30-34) - (55-64)   1.4408 0.0486 128  29.632 <0.0001
##  (30-34) - (65+)     2.0943 0.0486 128  43.071 <0.0001
##  (35-44) - (45-54)   0.5005 0.0486 128  10.294 <0.0001
##  (35-44) - (55-64)   1.0108 0.0486 128  20.788 <0.0001
##  (35-44) - (65+)     1.6642 0.0486 128  34.227 <0.0001
##  (45-54) - (55-64)   0.5103 0.0486 128  10.494 <0.0001
##  (45-54) - (65+)     1.1637 0.0486 128  23.933 <0.0001
##  (55-64) - (65+)     0.6534 0.0486 128  13.439 <0.0001
```

```
##
## sex = Male:
##  contrast          estimate     SE  df t.ratio p.value
##  (15-19) - (20-24)  -0.3173 0.0486 128  -6.525 <0.0001
##  (15-19) - (25-29)  -0.2414 0.0486 128  -4.965 <0.0001
##  (15-19) - (30-34)  -0.0373 0.0486 128  -0.766  0.9945
##  (15-19) - (35-44)   0.3310 0.0486 128   6.808 <0.0001
##  (15-19) - (45-54)   0.8412 0.0486 128  17.300 <0.0001
##  (15-19) - (55-64)   1.2691 0.0486 128  26.101 <0.0001
##  (15-19) - (65+)     1.7364 0.0486 128  35.711 <0.0001
##  (20-24) - (25-29)   0.0759 0.0486 128   1.560  0.7728
##  (20-24) - (30-34)   0.2800 0.0486 128   5.759 <0.0001
##  (20-24) - (35-44)   0.6483 0.0486 128  13.333 <0.0001
##  (20-24) - (45-54)   1.1585 0.0486 128  23.825 <0.0001
##  (20-24) - (55-64)   1.5864 0.0486 128  32.626 <0.0001
##  (20-24) - (65+)     2.0537 0.0486 128  42.236 <0.0001
##  (25-29) - (30-34)   0.2042 0.0486 128   4.199  0.0013
##  (25-29) - (35-44)   0.5725 0.0486 128  11.773 <0.0001
##  (25-29) - (45-54)   1.0826 0.0486 128  22.265 <0.0001
##  (25-29) - (55-64)   1.5105 0.0486 128  31.066 <0.0001
##  (25-29) - (65+)     1.9778 0.0486 128  40.676 <0.0001
##  (30-34) - (35-44)   0.3683 0.0486 128   7.574 <0.0001
##  (30-34) - (45-54)   0.8784 0.0486 128  18.066 <0.0001
##  (30-34) - (55-64)   1.3064 0.0486 128  26.867 <0.0001
##  (30-34) - (65+)     1.7737 0.0486 128  36.477 <0.0001
##  (35-44) - (45-54)   0.5101 0.0486 128  10.492 <0.0001
##  (35-44) - (55-64)   0.9381 0.0486 128  19.293 <0.0001
##  (35-44) - (65+)     1.4054 0.0486 128  28.903 <0.0001
##  (45-54) - (55-64)   0.4279 0.0486 128   8.801 <0.0001
##  (45-54) - (65+)     0.8952 0.0486 128  18.411 <0.0001
##  (55-64) - (65+)     0.4673 0.0486 128   9.610 <0.0001
##
## P value adjustment: tukey method for comparing a family of 8 estimates
```

2.3 Code

All data and scripts required to reproduce this report are available at:

GitHub: https://github.com/lydiamichael03/datanalysisR

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3 Results

3.1 Visualization

```
# Mean rate by age group and sex
mean_rates <- gonorrhea %>%
  group_by(sex, age_group) %>%
  summarise(mean_rate = mean(rate), se = sd(rate) / sqrt(n()), .groups = "drop")
```

```r
ggplot(gonorrhea, aes(x = age_group, y = rate, color = sex, group = sex)) +
  geom_jitter(alpha = 0.3, width = 0.15, size = 1.5) +
  geom_line(data = mean_rates, aes(y = mean_rate), linewidth = 1.1) +
  geom_point(data = mean_rates, aes(y = mean_rate), size = 3, shape = 21,
             fill = "white", stroke = 1.5) +
  scale_color_manual(values = c("Female" = "#E07B6A", "Male" = "#4A90D9")) +
  labs(
    title = "Gonorrhea Incidence Rate by Age Group and Sex (2000-2022)",
    x = "Age Group",
    y = "Rate per 100,000 Population",
    color = "Sex"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    axis.text.x = element_text(angle = 30, hjust = 1),
    legend.position = "top"
  )
```

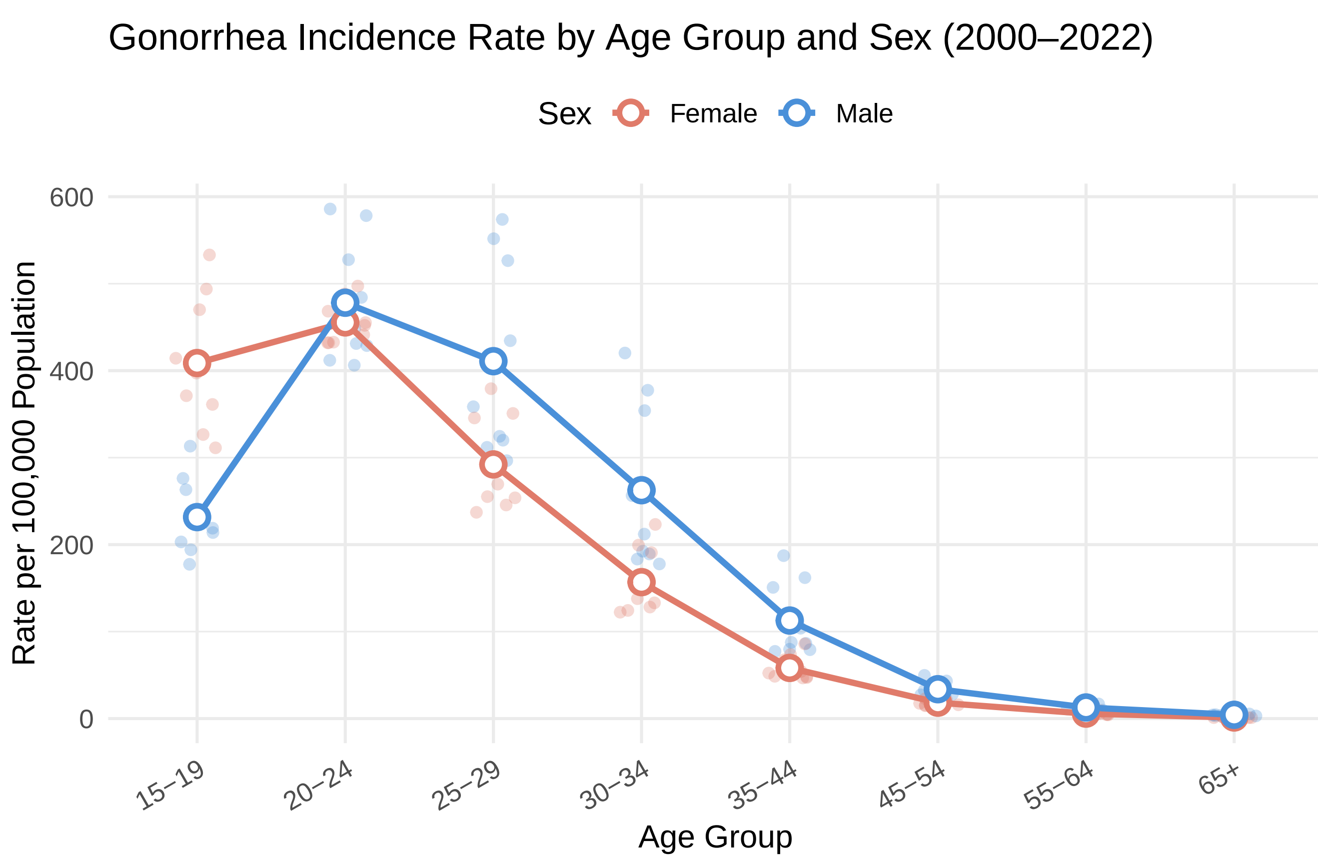

Figure 1: Gonorrhea incidence rate (per 100,000) by age group and sex, 2000–2022. Points show individual yearly observations; lines connect group means.

```r
ggplot(gonorrhea, aes(x = age_group, y = rate_log, fill = sex)) +
  geom_boxplot(alpha = 0.7, outlier.shape = 21, outlier.size = 2) +
  scale_fill_manual(values = c("Female" = "#E07B6A", "Male" = "#4A90D9")) +
```

```
  labs(
    title = "Log-Transformed Gonorrhea Rate by Age Group and Sex",
    x = "Age Group",
    y = expression(log[10](Rate~per~100000)),
    fill = "Sex"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    axis.text.x = element_text(angle = 30, hjust = 1),
    legend.position = "top"
  )
```

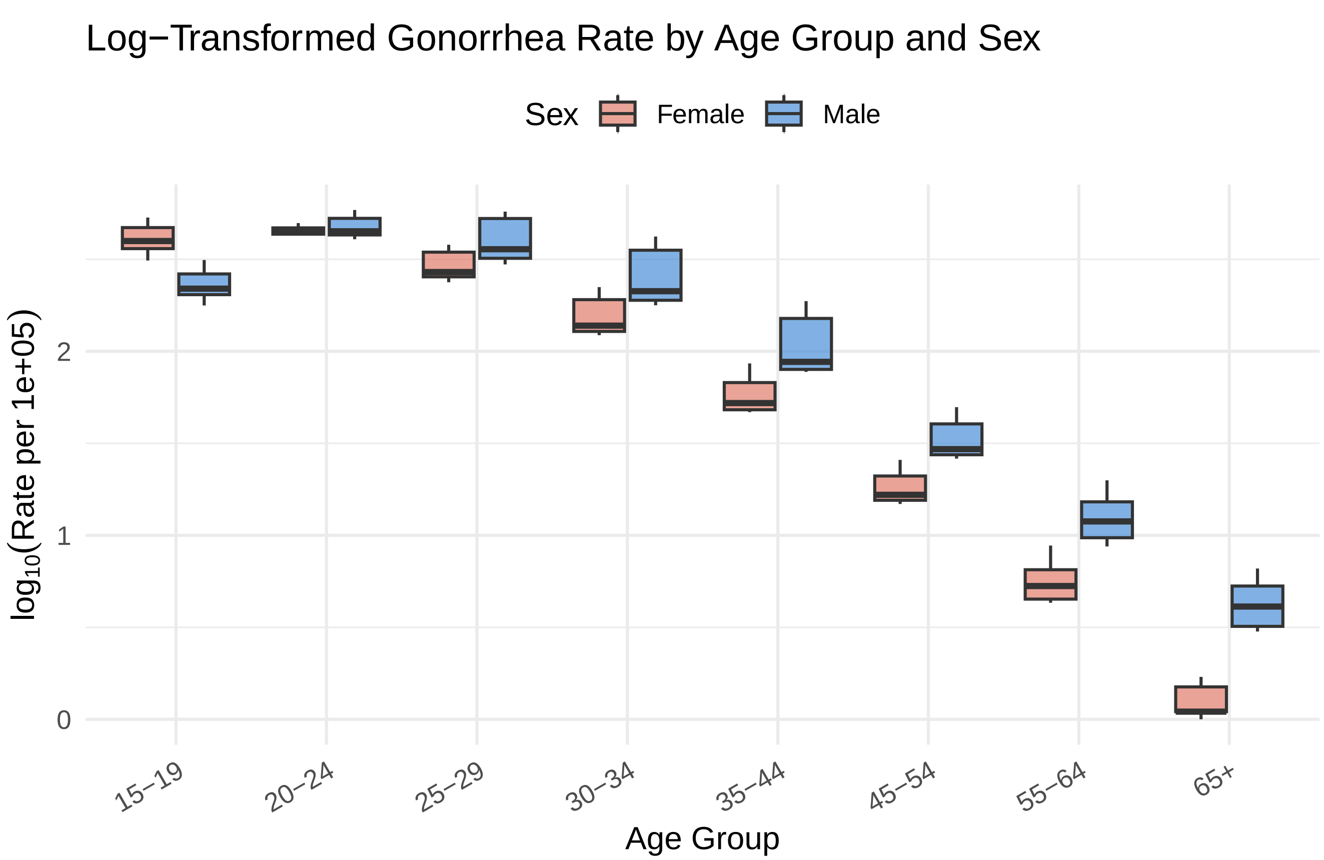

Figure 2: Distribution of log10-transformed gonorrhea rate by sex and age group. Boxes show median and IQR; whiskers extend to 1.5×IQR.

```
time_trends <- gonorrhea %>%
  mutate(year_num = as.integer(as.character(year))) %>%
  group_by(sex, year_num) %>%
  summarise(mean_rate = mean(rate), .groups = "drop")

ggplot(time_trends, aes(x = year_num, y = mean_rate, color = sex, group = sex)) +
  geom_line(linewidth = 1.1) +
  geom_point(size = 3) +
  scale_color_manual(values = c("Female" = "#E07B6A", "Male" = "#4A90D9")) +
```

```r
  scale_x_continuous(breaks = unique(time_trends$year_num)) +
  labs(
    title = "Mean Gonorrhea Rate Over Time by Sex (2000-2022)",
    x = "Year",
    y = "Mean Rate per 100,000 Population",
    color = "Sex"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    axis.text.x = element_text(angle = 30, hjust = 1),
    legend.position = "top"
  )
```

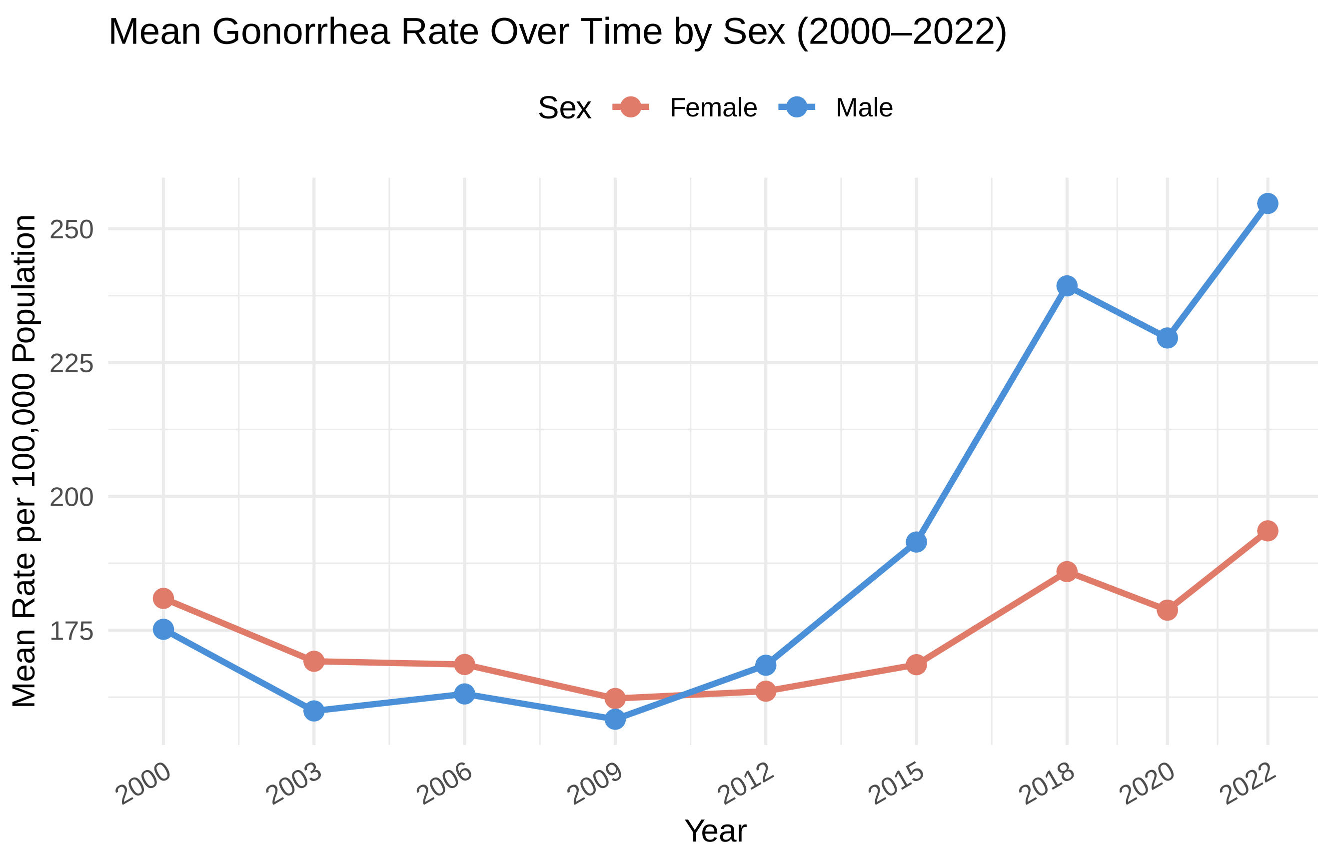

Figure 3: Trends in mean gonorrhea rate over time by sex, pooled across all age groups.

3.2 Statistical Analyses

The two-way ANOVA revealed statistically significant main effects of both sex and age group on $\log_{10}$-transformed gonorrhea rates, as well as a significant interaction between the two factors (all $p < 0.05$; see ANOVA table above).

Main effect of sex: Gonorrhea rates differed significantly between males and females overall. On average, females in younger age groups (15–19, 20–24) showed higher rates than males, while males showed higher rates in older age groups — a pattern consistent with the significant sex $\times$ age group interaction.

Main effect of age group: Rates varied substantially across age groups, with the highest rates consistently observed in the 15–24 year range and declining sharply with increasing age.

Interaction (sex × age group): The significant interaction indicates that the sex difference in rates is not uniform across all age groups. Post-hoc pairwise comparisons (Tukey-adjusted) identified which specific age group comparisons drove these differences within each sex.

---

4 Conclusions

The results demonstrate that gonorrhea incidence rates in the United States (2000–2022) are significantly associated with both sex and age group, with a meaningful interaction between the two. These findings are broadly consistent with existing CDC surveillance data and the published literature [@cdc2023; @leichliter2007].

The higher rates observed in adolescent females (15–19) relative to males of the same age align with the known biological vulnerability of younger females due to cervical ectopy [@haggerty2010]. The convergence or reversal of this sex differential in older age groups likely reflects the increasing contribution of MSM transmission to male rates in recent decades [@cdc2023].

The sharp decline in rates with age in both sexes underscores the concentration of gonorrhea transmission in younger populations and supports the continued prioritization of STI screening and sexual health education for adolescents and young adults. These findings reinforce current CDC screening recommendations, which target sexually active women under 25 and men at increased risk [@cdc2023].

Future analyses should incorporate data on race/ethnicity and geographic region, both of which are known to modify gonorrhea risk and could confound the sex- and age-based patterns observed here.

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5 AI Use Disclosure Statement

This document was produced with the assistance of Claude (Anthropic) to generate the initial R Markdown structure and code framework, which were then reviewed and adjusted by the author to ensure accuracy and alignment with the dataset.